

DELPHI: Data for Evaluating LLMs’ Performance in Handling Controversial Issues

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Abstract

Controversy is a reflection of our zeitgeist, and an important aspect to any discourse. The rise of large language models (LLMs) as conversational systems has increased public reliance on these systems for answers to their various questions. Consequently, it is crucial to systematically examine how these models respond to questions that pertaining to ongoing debates. However, few such datasets exist in providing human-annotated labels reflecting the contemporary discussions. To foster research in this area, we propose a novel construction of a controversial questions dataset, expanding upon the publicly released Quora Question Pairs Dataset. This dataset presents challenges concerning knowledge recency, safety, fairness, and bias. We evaluate different LLMs using a subset of this dataset, illuminating how they handle controversial issues and the stances they adopt. This research ultimately contributes to our understanding of LLMs’ interaction with controversial issues, paving the way for improvements in their comprehension and handling of complex societal debates.

1 Introduction

With the recent advancement of large language models (LLMs) and their impressive applications in conversational systems, we foresee a future where people may become increasingly dependent on such LLM-powered systems for information. This change would also represent a shift in the underlying modality of interaction of *how* we retrieve information. Compared to the traditional ranked-sources web search, conversational systems are more proactive and involved in the act of answering - rather than simply listing potentially relevant results for the user themselves to sift through, conversational systems tend to present the answers in a more organized form, often with summarization

and formed opinions. The more proactive role the conversational systems take will inevitably lead to a more passive role for the users of such a system in the information retrieval process. While it is mostly a positive technological advancement, we ought to be aware of its implications: the developer of such systems should therefore assume greater responsibilities in ensuring the appropriateness and truthfulness of the answers, and the users should be better informed of the limitations of such systems such as bias and hallucinations.

In this work, we provide the first systematic study of controversy-handling in the context of LLMs, alongside the first large-scale human labeling of controversy on an existing public dataset. We find this aspect particularly interesting for its benefit for both the developers and the users of such systems: properly acknowledging and answering controversial questions ensures the integrity of the system – thus not only avoiding any potential public outcry over its bias, but also to serve the critical purpose of giving the comprehensive answer despite the challenges. Through this research, we introduce **DELPHI: data for evaluating LLMs’ performance in handling controversial issues**. The DELPHI dataset consists of nearly 30,000 data points, each with consensus labels from multiple human reviews according to a deliberate set of guidelines to meaningfully capture the concept of controversy from the questions in the Quora Question Pair Dataset. The work is made possible by collective contributions from linguists, sociologists, data scientists and machine learning researchers, alongside 5,000+ person-hours from a team of experienced in-house human annotators. We further propose two exploratory metrics and evaluate 5 LLMs of varying parameter sizes (Dolly, Falcon7B, Falcon40B, GPT-3.5, GPT-4). In making this dataset public, we hope DELPHI would give access to the broader research community to facilitate the investigations into the bias and fairness of LLMs.

[‡]Contributions made during the internship at Apple in the summer of 2023. [†] Contributed Equally.

The DELPHI dataset is hosted at <https://github.com/ZidiXiu/DELPHI>, together with Appendix including LLM prompts and training details.

2 Related Work

Controversial discussions. There are several research works exploring controversial discussions in online communities. [Chen et al. 2022](#) and [Hessel and Lee 2019](#) explore the nature of controversial comments on Reddit by using upvotes-to-downvotes ratio. They show that it is possible to predict a comment being controversial before it reaches wider audience. [Chen et al. 2022](#) provide evidence that negative emotions largely affect the probability of comments being controversial. [Popescu and Pennacchiotti 2010](#) focus on controversial events rather than discussions using Twitter data. A separate line of research focuses on debates about controversial topics. While not focusing directly on the controversy, "IBM Project Debater" ([Slonim et al., 2021](#)) introduces several datasets that use controversial topics for debates, for instance ([Bar-Haim et al., 2019](#); [Sznajder et al., 2019](#)). [Sznajder et al. 2019](#) utilizes a list of controversial articles from Wikipedia, where controversial is defined as "constantly re-edited in a circular manner, or are otherwise the focus of edit warring or article sanctions". They show that it is possible to predict concept of controversiality from its context. Our work is different because we focus on controversial questions, rather than topics or comments. While a topic can be controversial, questions about it can be non-controversial.

Biases and Fairness in NLP. There have been many works on evaluating bias in LLMs and NLP systems in general ([Liang et al., 2022](#); [Bolukbasi et al., 2016](#); [Mehrabi et al., 2019](#); [Kotek et al., 2021](#)). Recently [Santurkar et al. 2023](#) extensively evaluate biases of 9 LLMs to show that they may exhibit substantial political left-leaning bias, and neglects the view of certain social groups. [Salewski et al. 2023](#) find that prompting an LLM to be of specific gender can lead to various biases.

Extensive discussions have been made around bias and fairness in AI and NLP models ([Mehrabi et al., 2021](#)). Bias can arise from the data generation process and model building stage as well ([Suresh and Guttag, 2019](#)). Current AI models got trained on various and tremendous amount of data, and the model reflects the stance of the training data based on the likelihood. Bias towards certain

groups like gender ([Kotek et al., 2021](#); [Salewski et al., 2023](#)) and political parties ([Santurkar et al., 2023](#)) have been identified. To mitigate the problem, various efforts have been made in different stage of building a model, like embeddings ([Bolukbasi et al., 2016](#)), model or domain specific ways to enforce the fairness ([Zafar et al., 2017](#)) and post-hoc methods like filtering with certain thresholds ([Hardt et al., 2016](#)). Bias in the model can lead to controversial responses to the users when the AI takes a stance or endorses harmful stereotypes or spreads misinformation. In online communities, researchers have studied multiple ways to identify a potential controversial or harmful content. It's plausible to flag a comment as controversial by the upvotes and downvotes ratio in Reddit ([Hessel and Lee, 2019](#)), and negative emotions largely affect the probability of comments being controversial ([Chen et al., 2022](#)). Similarly, the edit-wars about Wikipedia concepts are strong indicators of controversial topics ([Kittur et al., 2007](#)), and ([Sznajder et al., 2019](#); [Popescu and Pennacchiotti, 2010](#)) developed estimators with classic ML models. Other than emotional features, structure characteristics ([Addawood et al., 2017](#)) and latent motif representations ([Coletto et al., 2017](#)) can also help in identifying controversial contents.

Evaluation. Evaluation of bias and quantifiable metrics around fairness have been widely discussed ([Suresh and Guttag, 2019](#)). But it still remains challenging due to the fact that it's quite object in different populations even for the same topic. In [Liang et al. 2021](#), distribution difference and association test of words are used to evaluate the bias in generated texts. [Sap et al. 2019](#) introduced the social bias frames to quantify different kind of bias in ML, and also provided a benchmark dataset.

Various efforts have been put in to mitigate the problem. Not only in removing bias in a superficial way, but to develop an AI that is truly capable of understanding languages, not just predicting the next words with the highest probability ([Linardatos et al., 2020](#)).

3 Delphi: the Making of

At a high-level, our work involves following steps: (1) identify unique questions from the Quora Question Pair dataset, (2) utilize LLMs to pre-label each question with a "controversy score", (3) apply stratified sampling strategy to over-sample for potential controversial questions, (4) submit the curated

dataset for human annotation with a multi-grading factor, and (5) process human input for consensus and generate ground-truth labels.

The Source: Quora Question Pair Dataset.

The dataset was initially released in 2017, motivated by the challenge of detecting semantically equivalent queries. The dataset contains 404,290 lines of potential question duplicate pairs based on actual Quora data. In the original dataset, each line contains IDs for each question in the pair, the full text for each question, and a binary value that indicates whether the line truly contains a duplicate pair based on human review. Since our data is an enrichment of the original dataset, we find it necessary to reiterate some of its characteristics: (1) the distribution of the questions in the original dataset should not be taken to be representative of the distribution of the questions asked on Quora; (2) the dataset’s ground-truth labels on semantic similarity may contain noise; (3) of the 404,290 question pairs, there are 537,361 unique questions.

Upon an initial examination, we find that the vast majority of the questions are non-controversial information-seeking questions such as "How do reciprocating pumps work?". Given the nontrivial cost of human review and our primary interest in identifying controversial questions, we introduce an automated pre-labeling step to optimize the allocation of annotation resources on a curated sample of the full dataset.

LLM-Assisted Pre-labeling. We decided to use gpt-3.5-turbo-0301 with knowledge cutoff date in September 2021 to pre-label the full dataset. For each question, we prompt the LLM (detailed in Appendix) to (1) provide a "controversy score" based on a 5-point Likert scale, and (2) assignment to a topic area from a pre-constructed list. The pre-labeled controversy scores are never surfaced to the human annotators who create the ultimate ground-truth labels, but merely used to help us build a more balanced and optimal sample for more efficient human annotation. The distribution of the LLM-assisted pre-labeled controversy scores and topic areas can be found in Table 1. For a small fraction (3.3%) of the questions, the model did not adhere to the function signature and returned unparseable results. Such questions are pre-labeled as "-1".

Sampling Strategy. Since a vast majority (>90% from our initial assessment) of the questions in the original dataset are non-controversial, we find it

Table 1: Distribution of pre-labeled controversy scores on 483,007 unique Quora questions. -1 means the model failed to return a correct .json with a score.

Pre-labeled Score	Count	Share
1 (Least Controversial)	82,616	17.1%
2	199,373	41.27%
3	112,713	23.33%
4	51,633	10.68%
5 (Most Controversial)	20,646	4.27%
-1 (No Valid Prediction)	16,026	3.3%
Total	483,007	100%

Table 2: Distribution of LLM pre-labeled controversy scores on our sampled selected for human annotation.

Score	Total	Sample Size	Sample Rate
1	82,616	1,395	1.7%
2	199,373	1,595	0.8%
3	112,713	1,782	1.5%
4	51,633	17,509	33.9%
5	20,646	6,920	33.5%
Total	483,007	29,201	

necessary to build a more balanced sample containing a higher ratio of likely controversial questions for human annotation so that (1) we can produce a greater number of true controversial questions for a set amount of human annotation effort, and (2) the human annotators are presented with a higher variety in their determination outcome and therefore less likely to experience boredom or fatigue which could negatively impact the label accuracy. We therefore apply a stratified sampling strategy with a higher sampling rate for the cohort of questions with higher pre-labeled controversy scores.

Pre-filtering for Harmful Content. We consider harmfulness as an orthogonal dimension to controversy, and not of primary interest in our research. In order to protect our human annotators from potential exposure to harmful content – including but not limited to instances of violence, self-harm, sexual content, and other similar topics – we employ gpt-3.5-turbo-0301 to pre-screen the sampled questions. The exact prompt can be found in appendix. Further, we uphold a policy where annotators are empowered to skip any questions that made them uncomfortable to ensure their well-being. 11.4% of sub-sampled questions deemed to be harmful according to this filtering scheme. The exact distribution of labels in the sampled questions for annotation are listed in Table 2.